

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 - CONCEPT MAPPING

Course Code:	CSA1205	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	C01	Assessment:	ASSESSMENT TOOL 2- CONCEPT MAPPING
Weightage:	35%	Total Marks:	25
C01 Statement:	Analyze the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking (BL4)		
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Department:	Cse-AI	Date:	

Instruction to Students

1. Answer two questions as per the Instruction.
2. Each question carries 12.5 mark.
3. Only the appropriate Tools can be used
4. Time allotted for the exam is 60 min

ANNEXURE A

1. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.
 2. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.
 3. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.
 4. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show real-world scenarios where specific addressing modes are preferred.
 5. Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system design.
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Code of Conduct:

I THIRUMALAPUDI PRANAVI 192411222(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

T.Pranavi

Signature of the student:

CO1-ASSESSMENT TOOL-2

U.Prabhavathi(192472067)

CSA1206

1. Concept Map: CPU, Memory and I/O Devices

Explanation

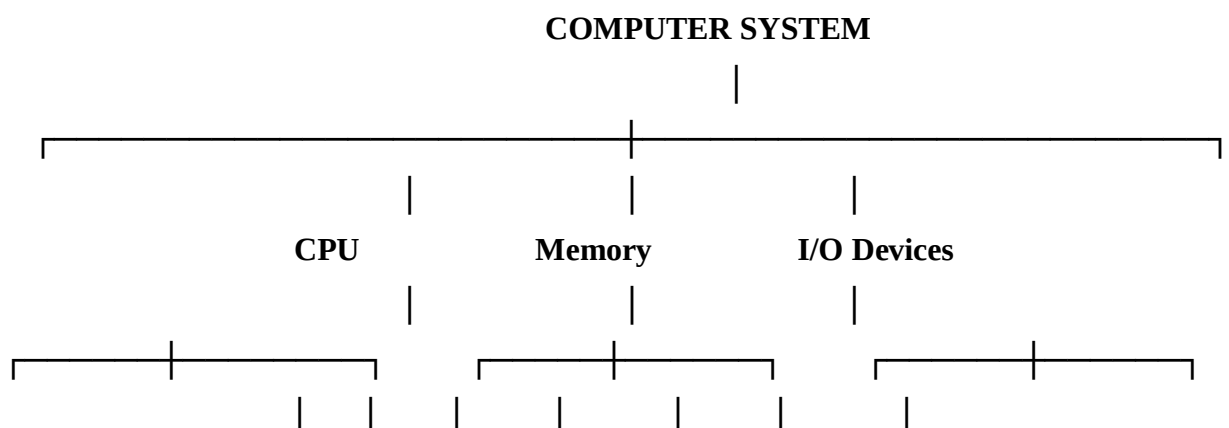
The CPU (Central Processing Unit) is the brain of the computer. It consists of the Arithmetic Logic Unit (ALU), Control Unit (CU), and Registers. The ALU performs arithmetic and logical operations, the Control Unit controls the execution of instructions, and Registers store temporary data for fast processing.

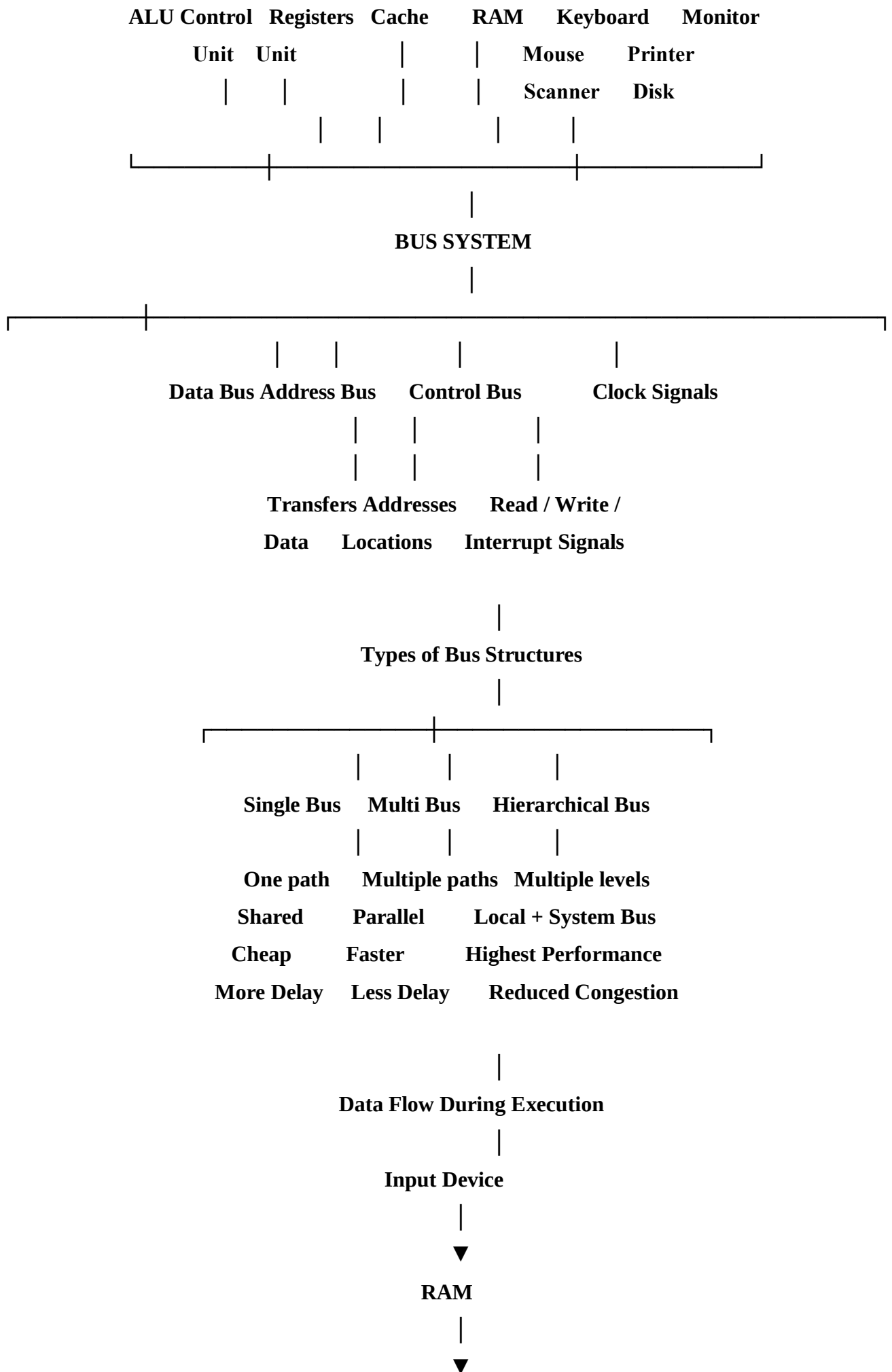
The Memory consists of Cache and RAM. Cache memory is very fast and stores frequently used data, while RAM stores programs and data currently being executed. The CPU communicates with memory through buses.

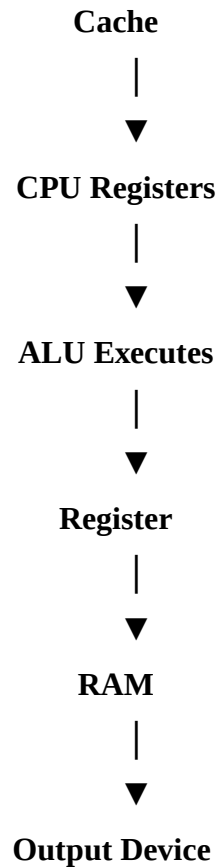
I/O devices such as keyboards, cameras, sensors, printers, and monitors provide input to the computer and display the output. Data and control signals travel through the system bus.

There are three main bus structures:

- Single Bus: CPU, memory, and I/O share one bus, causing congestion.
- Multi-Bus: Separate buses allow multiple data transfers simultaneously.
- Hierarchical Bus: Uses high-speed local buses connected to slower expansion buses for better performance.







2. Concept Map: Instruction Execution Cycle and CPU Performance

Explanation

The instruction execution cycle consists of Fetch, Decode, and Execute stages.

- Fetch: The CPU retrieves the instruction from memory.
- Decode: The Control Unit interprets the instruction.
- Execute: The ALU performs the required operation.

CPU performance depends on:

- Instruction Count (IC)
- Cycles Per Instruction (CPI)
- Clock Cycle Time

The execution time is calculated using:

$$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Cycle Time}$$

Memory delays, cache misses, branch instructions, and pipeline stalls increase CPI, which slows the processor.

Modern CPUs improve performance using:

- Pipeline execution
- Cache memory
- Branch prediction
- Multi-core processors

Benchmark programs are used to compare the performance of different processors.

3. Concept Map: 8085 vs 8086 Microprocessor

Explanation

The 8085 is an 8-bit microprocessor with a 64 KB memory space, whereas the 8086 is a 16-bit microprocessor capable of addressing 1 MB of memory.

The 8085 supports basic addressing modes such as:

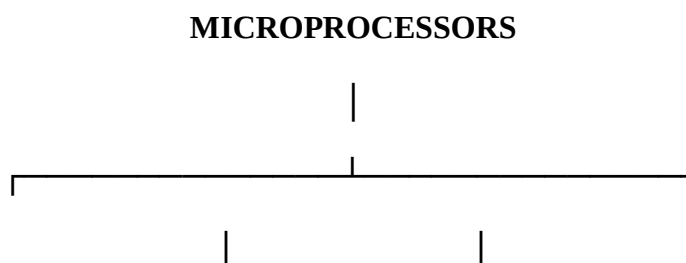
- Immediate
- Register
- Direct
- Indirect

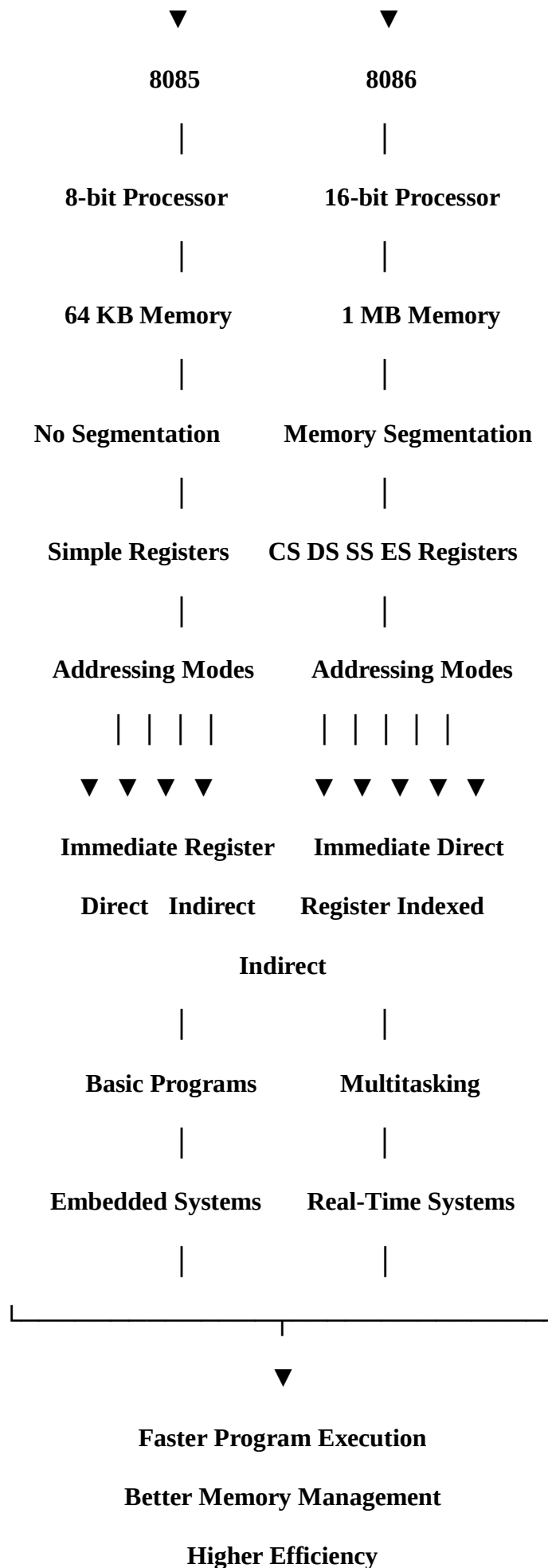
The 8086 additionally supports:

- Indexed
- Based
- Relative addressing modes

The 8086 introduces memory segmentation using:

- Code Segment (CS)
- Data Segment (DS)
- Stack Segment (SS)
- Extra Segment (ES)





4. Concept Map: Addressing Modes and Instruction Types

Explanation

Addressing modes specify how the CPU accesses data required by an instruction.

The common addressing modes are:

- Immediate: Data is part of the instruction.
- Register: Data is stored in CPU registers.
- Direct: Memory address is directly specified.
- Indirect: Address is stored in another register.
- Indexed: Effective address is calculated using an index register.

These addressing modes are used with different instruction types:

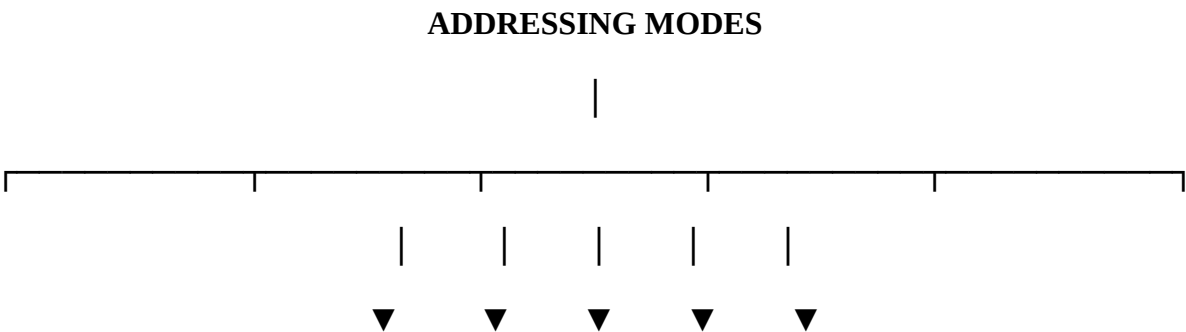
- Data Transfer Instructions: MOV, LDA, STA
- Arithmetic Instructions: ADD, SUB, INC
- Control Instructions: JMP, CALL, RET

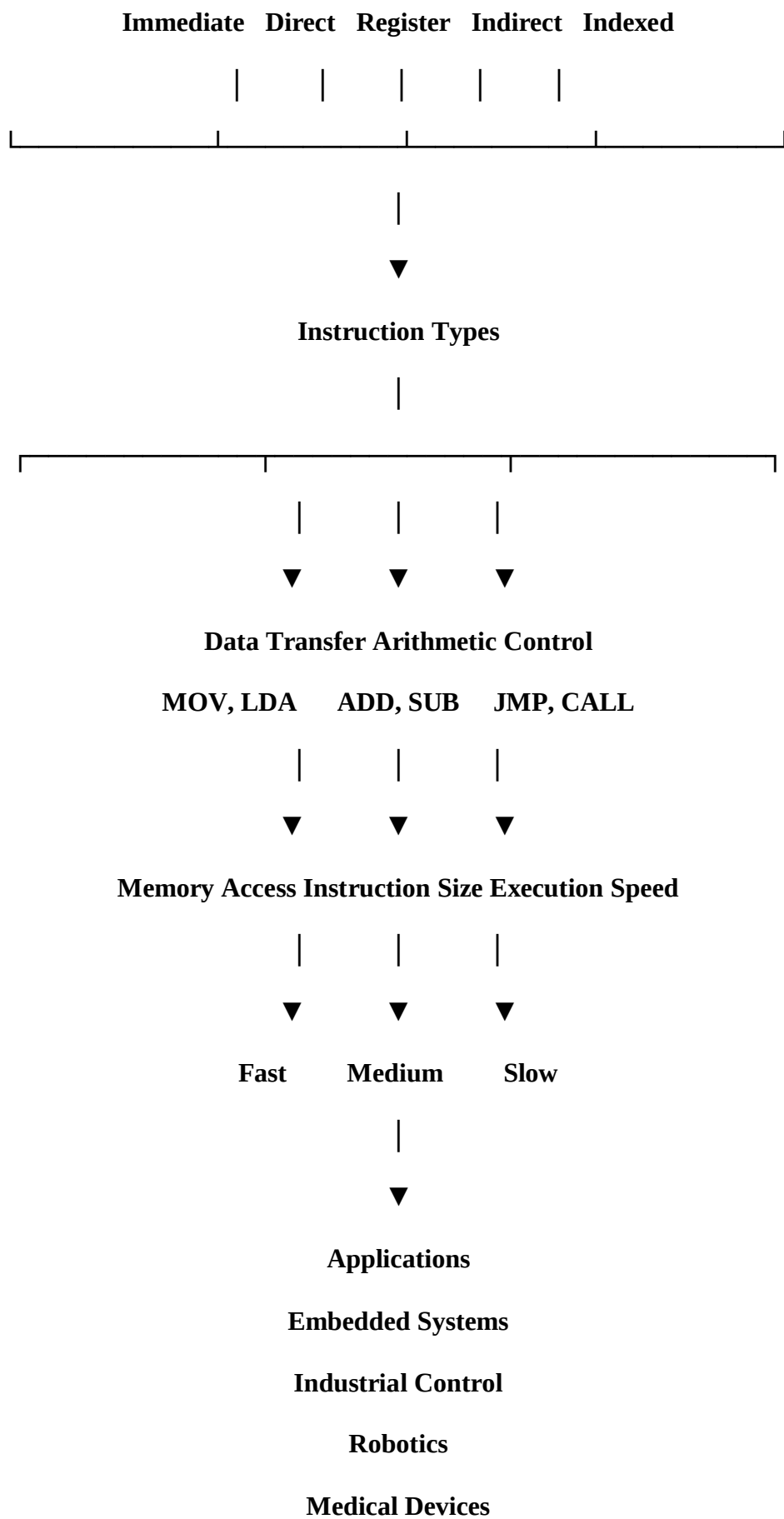
Each addressing mode affects:

- Memory access time
- Instruction size
- Execution speed

For example:

- Register addressing is the fastest because data is already inside the CPU.
- Direct and indirect addressing require memory access, making them slower.
- Indexed addressing is widely used in arrays and string processing.





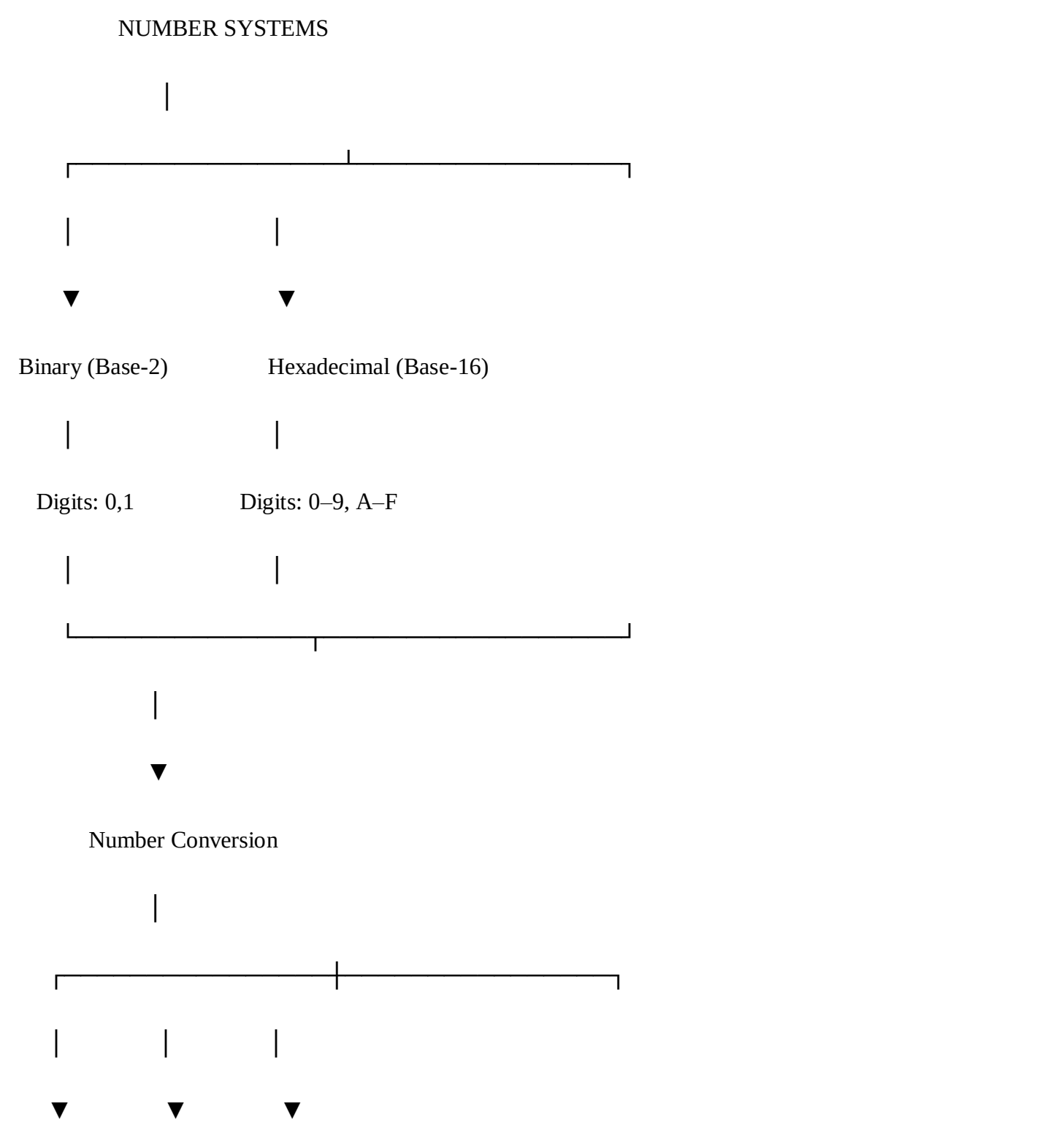
5. Concept Map: Number Systems and CPU Execution

Explanation

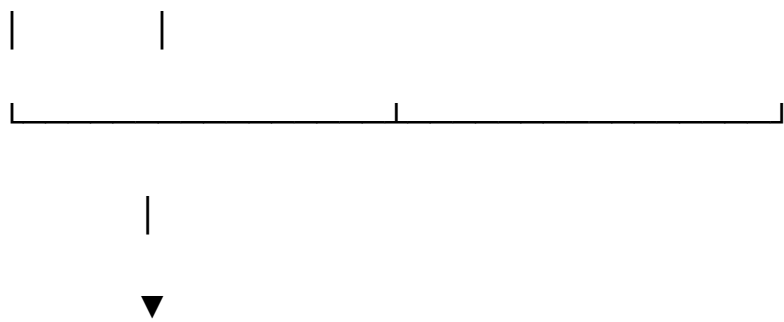
Computers internally use the binary number system (0 and 1) because digital circuits have only two states: ON and OFF.

Programmers prefer the hexadecimal number system because it is shorter and easier to read.

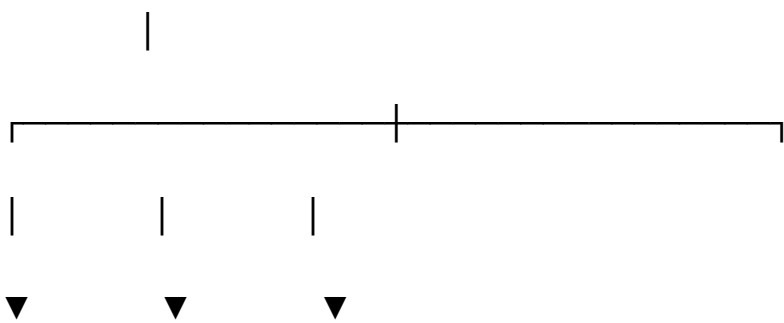
One hexadecimal digit represents 4 binary bits.



Binary → Hex Hex → Binary Decimal Conversion



Instruction Representation



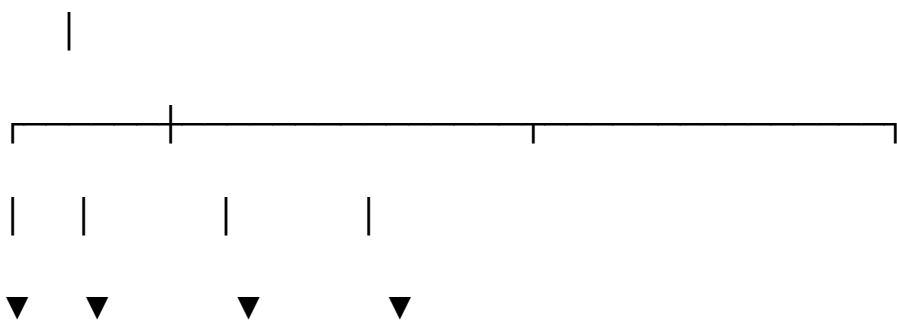
Opcode Field Operand Field Machine Code



Binary Instructions



Memory Storage



RAM Cache Main Memory Memory Address

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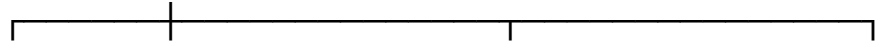
Instructions Stored as Binary Bits

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CPU Interpretation

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Fetch Decode Execute Store Result

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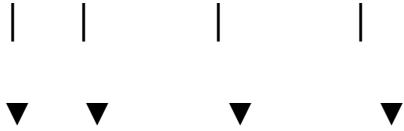
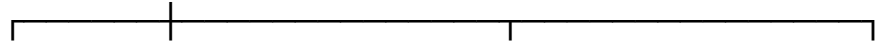
Arithmetic & Logical Operations

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Debugging Process

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Memory Dump Registers Machine Code Error Analysis

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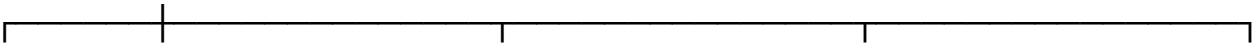
Uses Hexadecimal Values

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Advantages of Hexadecimal

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Compact Easy to Read Easy Debugging Memory Addresses

Representation Faster Analysis Used in Programming

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Low-Level Programming &

System Design

MARKS ALLOCATION

Criteria	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)
Q1					
Q2					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand